Simulink Model for Real-Time Logging of Process Data in MATLAB Workspace

In this project, temperature data is logged in real time in the MATLAB workspace, by building a model that uses Arduino IO Library in Simulink.

Process parameter measurement carried out for experimental research and analytical testing generally needs to be saved in real time in order to conduct further analysis.

This is particularly relevant in the domain of instrumentation engineering where parameters such as temperature, pressure, flow rate, pH, viscosity, etc, are measured and then saved on the computer in order to analyse the dynamics of the process, or for fault diagnosis. This project is based on Arduino Uno and temperature sensor LM35 IC. The author's prototype is shown in Figure 1.

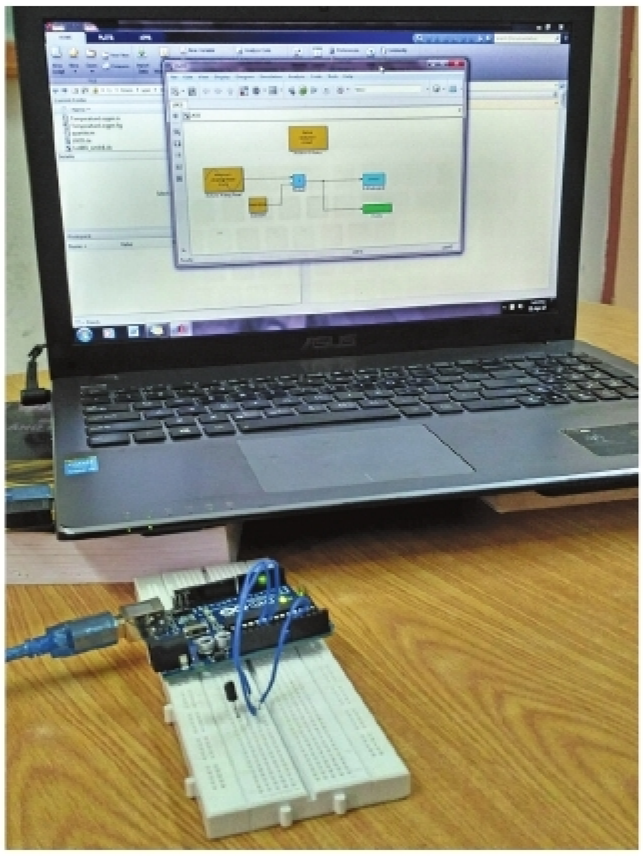

Figure 1: Author's prototype wired on breadboard

Hardware and construction

The components used in this project are:

Arduino Uno: Arduino Uno is an AVR ATmega328P microcontroller (MCU) based development board with six analogue input pins and 14 digital I/O pins. The MCU features 32kB ISP flash memory, 2kB RAM, and 1kB EEPROM. The board provides the capability of serial communication via UART, SPI, and I2C. It can operate at a clock frequency of 16MHz. In this project, the analogue pin A0 of Arduino is used to read the output voltage (from pin no. 2) of LM35.

LM35: This is a precision IC temperature sensor. Its output voltage is linearly proportional to the temperature (in degree Celsius). LM35 exhibits typical accuracies of $\pm 1/4$°C at room temperature and $\pm 3/4$°C over a full temperature range from –55°C to +150°C. This sensor provides a sensitivity of 10mV/°C. In this project, the Vout pin (pin no. 2) of LM35 IC is connected to analogue input pin A0 of Arduino. Pin numbers 1 and 3 of LM35 are connected to +5V and GND of the Arduino Uno board, respectively.

The connection diagram of the interfacing temperature sensor with the Arduino board is shown in Figure 2.

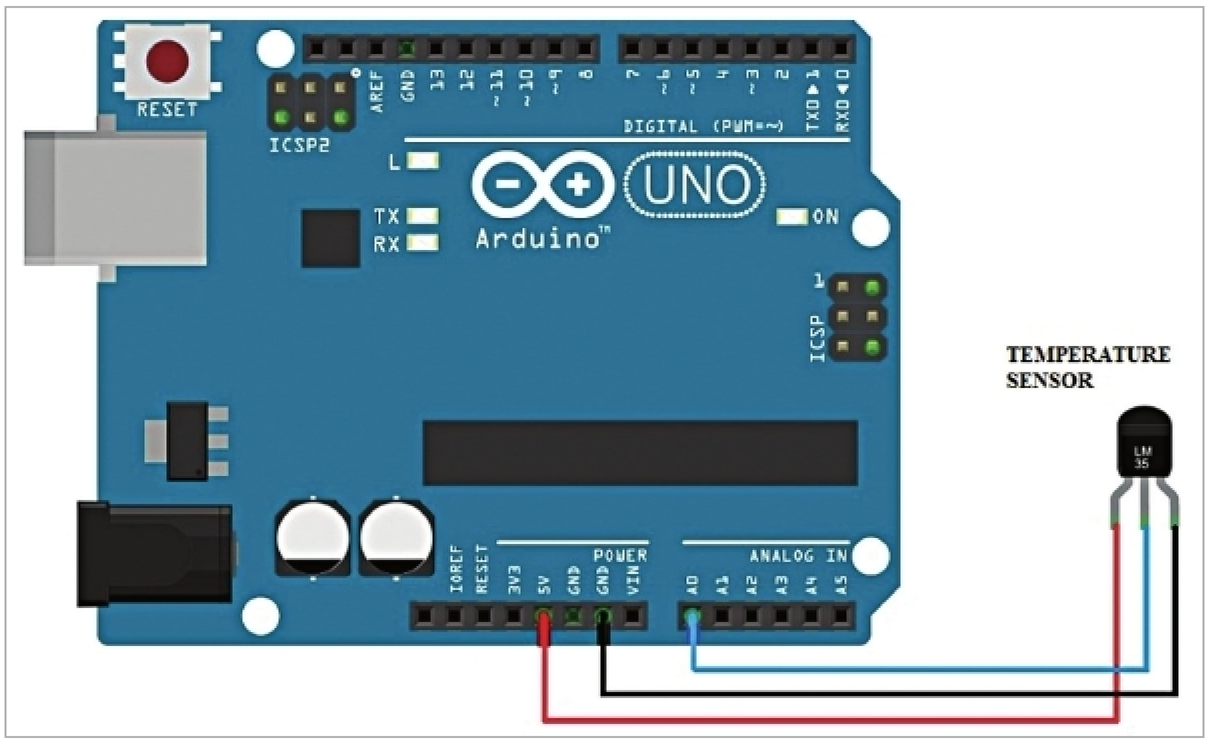

Figure 2: Connection diagram